

Josh Wang

Software Developer

Full-stack engineer specializing in React / Next.js front-end architecture, with Node.js backend and AI-assisted development workflow integration experience

Taipei, Taiwan | spjay1@gmail.com | profile.joshwang.dev

Summary

10+ years of full-stack development experience (primarily front-end), spanning SaaS, cybersecurity, cross-border internal systems, and React Native.

Led frontend architecture and delivery for 7+ web systems and 2 mobile apps using TypeScript, CI/CD, monorepo, and AWS.

Focused on scalable frontend systems, AI-assisted development workflows, and legacy system modernization.

Key Achievements

- Reduced cloud infrastructure costs by 90%
- AI-assisted POC validation: weeks to 1.5 days
- Supported internal systems across 50+ countries
- Designed a job-queue-driven LLM pipeline with self-healing retry logic

Projects

PickleScout — LLM-Driven E2E Test Generation Agent May 2026

Open Source Side Project | github.com/iskWang/PickleScout

- Completed full POC in one week using AI-assisted workflow; open-sourced with browser agent, LLM pipeline, self-healing, and full-stack architecture
- Designed two-pass LLM pipeline: Stagehand browser agent explores target app and records ActionLog, converted to Gherkin feature files and Playwright step definitions
- Implemented self-healing mechanism: on test failure, LLM auto-patches selectors and timeouts and retries, improving generated test usability
- Backend job queue (Fastify + BullMQ/Redis) decouples long-running LLM generation from requests, with failure retry and status tracking; generated output includes a GitHub Actions workflow with zero LLM dependency at runtime

Experience

Full-Stack Engineer (Contract-based) Jun 2023 – Present

Independent

- CI/CD Architecture & Security | Data Analytics SaaS Startup
- Led CI/CD workflow restructuring, consolidating 5 fragmented branches into 2 and optimizing the monorepo structure to reduce operational overhead
 - Introduced Claude + Playwright MCP for automated Gherkin script generation, reducing regression risk and improving release stability
 - Performed critical security patching under time pressure, resolving CVE-2025-29927 and strengthening system security posture
- Map System Development & Technical Debt Cleanup | Major Real Estate Group
- Served as technical lead to fill development gaps, delivering complex map POI filtering features on tight deadlines
 - Rolled out a Claude Code + MCP-assisted development workflow on the existing codebase, cutting feature POC validation time from weeks to 1.5 days
 - Led systematic module decomposition and refactoring to resolve legacy architectural debt at the root
- Real-Time Trading Feature Development | Vue 3 + Nuxt.js
- Took over an existing Vue 3 + Nuxt 3 + Pinia project, extending real-time trading features within its established architecture conventions
 - Integrated TradingView Widget and ECharts for real-time price feeds, order book visualization, and data analytics

Front-End Developer (Platform & Architecture)

Oct 2021 – Jun 2023

TeamT5 (Cybersecurity)

ThreatSonar Product — Frontend Architecture Planning

- Led frontend architecture planning for the frontend/backend separation, producing technical specs and architecture design docs as the implementation foundation

Official Website & Event Pages (CYBERSEC 2022/2023, InterOp23)

- Led cloud architecture migration from Firebase/Cloudinary to AWS S3/CloudFront + Next.js, reducing cloud costs by 90% and improving reliability
- Refactored 50%+ of the legacy codebase and implemented an AWS SDK/TypeScript/Pica.js file-upload system with auto-thumbnail generation
- Built CYBERSEC 2022/2023 and InterOp23 event pages with Preact.js + Tailwind CSS, deployed via AWS S3 + CloudFront
- Identified plaintext password transmission and auth design flaws in an external vendor system; escalated and drove remediation

Front-End Lead

May 2019 – Oct 2021

OEC Group

B2B/B2C Back-office System

- Led frontend rebuild using React Hooks and Context API with real-time WebSocket messaging, covering internal systems across 50+ countries
- Designed 100-field single-page forms using Formik / Yup / React.memo with strict validation and performance control
- Proposed and adopted GitHub Actions CI/CD pipeline as an early adopter upon its launch in Aug 2019

Website Redesign / Internal Training

- Took over external vendor's HTML/CSS and completed Next.js + i18n architecture (13 languages) within one month
- Led Pull Request Code Review culture to improve code quality and ensure requirement alignment

Web Developer

Jul 2014 – Apr 2019

Startups (Fullinn, CatFi, myTreat, Ecowork)

Full-stack Architecture & API Design | Fullinn, myTreat, Ecowork

- Led frontend architecture with Next.js SSR to build an SEO-optimized hotel booking website builder; first client ranked on Google's first page within one week of launch
- Built OTA platform and order system with React.js / GraphQL, and led initial REST API design conventions to streamline backend development
- Delivered full-stack features and a standalone customer service back-office using Rails and Angular.js

Mobile & Cross-platform Development | Fullinn, CatFi

- Built cross-platform mobile apps with React Native, Redux, and Redux-saga for state and side effect management
- Integrated Firebase Push Notification and Code Push to reduce deploy costs and streamline push services
- Adopted Fabric and OneSignal for crash reporting and push; integrated WebRTC / RTMP streaming

Testing & Dev Workflow | Ecowork

- Implemented Cucumber and RSpec to fulfill client test report requirements; assisted DevOps in building automated test environments
- Introduced Vagrant-based deployment workflow to unify dev environments for faster onboarding

Education

Wenzao Ursuline University of Languages

Sep 2009 – Jun 2013

B.A. Digital Content Application and Management

Skills

Frontend	React / Next.js / TypeScript / Redux / Tailwind CSS / Vue 3 / Nuxt.js
Backend	Node.js / Fastify / Rails / Koa / RESTful API / GraphQL / Redis / BullMQ
Architecture & Infra	Monorepo / CI/CD / GitHub Actions / GitLab Runner / Docker / AWS
AI Workflow	Claude Code / MCP / Windsurf
Testing	Playwright / Jest / Cucumber / RSpec
Mobile	React Native / TestFlight